

Attachment #1: Table 1 Chemical Feed Pumps

Tag	Name	Concentration	Control	Pump Manufacturer	Model No	Flow	Head	HP	Speed	PH	HZ	Volt
02-01-001	Bleach UF Break Tank Feed - Biocontrol, Pump No. 1	12.5%	Analogue Or Pulse control	Prominent	GMXA1604PVT2M000UDC0300EN	0.95 gph	232 psig	30W	N/A	1	60	120
02-01-025	Bleach Pre - UF Feed - Biocontrol, Pump No. 2	12.5%	Analogue Or Pulse control	Prominent	GMXA1604PVT2M000UDC0300EN	0.95 gph	232 psig	30W	N/A	1	60	120
02-01-003	Bleach CIP, Pump No. 1	12.5%	On/off	Finish Thompson	DB9V-E-FF-M219	17.1 gpm	25 psig	1.5	3450	3	60	460
02-01-005	Citric Acid for CIP, Pump No. 1	50%	On/off	Finish Thompson	DB9V-E-FF-M219	17.1 gpm	25 psig	1.5	3450	3	60	460
02-07-002	Sulfuric Acid for UF CIP and waste neutralization, Pump No. 1	93%-98%	Analogue Or Pulse control	Prominent	S2CBH07220PVT5070UD11001EN	71.6 gpm	101 psig	110 W	N/A	1	60	120
02-01-009	Bisulfite for chlorine neutralization prior to RO, Pump No. 1	38%	Analogue Or Pulse control	Prominent	GMXA1604PVT2M000UDC0300EN	0.95 gph	232 psig	30W	N/A	1	60	120
02-01-011	Bisulfite for CIP, Pump No. 1	38%	On/off	Finish Thompson	DB9V-E-FF-M219	17.1 gpm	25 psig	1.5	3450	3	60	460
02-01-013	Anti-scalent prior to SWRO, Pump No. 1	100%	Analogue Or Pulse control	Prominent	GMXA1604PVT2M000UDC0300EN	0.95 gph	232 psig	30W	N/A	1	60	120
02-01-015	RO Cleaning Chemical Kleen MCT 113, Pump No. 1	100%	On/off	Finish Thompson	DB9V-E-FF-M219	17.1 gpm	25 psig	1.5	3450	3	60	460
02-01-017	RO Cleaning Chemical Kleen MCT 405, Pump No. 1	100%	On/off	Finish Thompson	DB9V-E-FF-M219	17.1 gpm	25 psig	1.5	3450	3	60	460
02-01-019	pH adjustment post RO, Pump No. 1	25%	Analogue Or Pulse control	Prominent	GMXA1604PVT2M000UDC0300EN	0.95 gph	232 psig	30W	N/A	1	60	120
02-01-021	pH adjustment CIP, Pump No. 1	25%	On/off	Finish Thompson	DB9V-E-FF-M219	17.1 gpm	25 psig	1.5	3450	3	60	460

Attachment #2: Venting Requirements for Polyethylene Tanks

Venting Requirements For Polyethylene Tanks									
Mechanical Pump Fill	Pneumatic Fill								
IF ≤ 1000 gallons	IF - Vent length ≤ 3 feet			IF - Vent length > 3' and ≤ 30'			IF - Scrubber Application		
Vent size should equal size of largest fill or discharge fitting	AND - Vent screen mesh size ≥ 1/4" or no screen used			AND - 3 or less 90° elbows with no other restrictions or reduction in pipe size			Vent pipe size throughout scrubber system CANNOT be reduced!		
							Centerline of dispersion pipe not to be submersed > 6 inches		
IF > 1000 gallons	Emergency Pressure Relief Cover Required			Emergency Pressure Relief Cover Required			Perforated dispersion pipe must be same diameter or larger, as vent. Sum of perforations ≥ cross sectional area of pipe		
Vent size should exceed the largest fill or discharge fitting by 1 inch	Tanker Discharge	Inlet/Fitting Size	Minimum Vent Size	Tanker Discharge	Inlet/Fitting Size	Minimum Vent Size	Tanker Discharge	Inlet/Fitting Size	Minimum Vent Size
	2"	2"	4"	2"	2"	6"	2"	2"	6"
	3"	2"	6"	3"	2"	6"	3"	2"	8"
	3"	3"	6"	3"	3"	8"	3"	3"	10"

(2) 2 inch vents **DO NOT EQUAL** 4 inch venting capacity

For detailed venting guidelines, please visit our Technical Resources at www.polyprocessing.com

rev. Nov 2006

Attachment #3: Table 2 Chemical Tanks

Tag	Name	Concentration	Style	Size, gal	Dia, ft	H, ft	Ladder	Linear PE Liner	Manway, in	Vent, in	Level Element opening, in	PRV Return Line, in
05-05-002	UF Break Tank	Seawater - 33 ppth	Single Wall	20,000	13'-9"	20'-3"	Yes	No	24	8	3 -Flanged 1-Flanged backup	N/A
05-07-003	CIP Tank	All Chemicals below except Anti-scalent. Max. concentrations in the tank will be as follows (not concurrently): 1. 500 mg/L as sodium hypochlorite 2. 2 g/L citric acid adjust down to pH 2 with Sulfuric Acid 3. 1.4 g/L as sodium bisulfite 4. Caustic Sode to neutralize citric acid. 5. High and low pH cleaner for RO (MCT 405 and MCT 113). Roughly 40 gallons into 2,700 gals of permeate water.	Double Walled	5,000	8'-6"	17'-11"	No	No	24	8	3 -Flanged 1-Flanged backup	- 3 inch Permeate water fill
02-07-001	Citric Acid	50% dry wt/wet wt	Double Walled	1,000	6'-5"	7'-1"	No	No	17	6	3 -Flanged 1-Flanged backup	1
02-07-002	Sulfuric Acid	93% dry wt/wet wt	Double Walled	405	4'-0"	6'-1"	No	Yes	7 threaded	4	3 -Flanged 1-Flanged backup	1
02-07-003	Sodium Bisulfite	38% dry wt/wet wt	Double Walled	1,000	6'-5"	7'-1"	No	No	17	6	3 -Flanged 1-Flanged backup	1
02-07-004	Caustic Soda	25% dry wt/wet wt	Double Walled	1,000	6'-5"	7'-1"	No	No	17	6	3 -Flanged 1-Flanged backup	1
02-07-005	Antiscalent HYPERSPERSE MDC714	100% wet weight	Double Walled	1,000	6'-5"	7'-1"	No	No	17	6	3 -Flanged 1-Flanged backup	1
02-07-006	Kleen MCT 405	These are all wet wt./wet wt. 1-3% Benzenesulfonic acid 3-10% Sodium alkyl benzene sulphonate 3-10% Sodium carbonate 3-10% N-hydroxyethylenediamine triacetic acid trisodium salt	Double Walled	405	4'-0"	6'-1"	No	No	7 threaded	4	3 -Flanged 1-Flanged backup	1
02-07-007	Sodium Hypochlorite	12.5% trade weight	Double Walled	1,000	6'-5"	7'-1"	No	Yes	17	6	3 -Flanged 1-Flanged backup	1
02-07-008	Kleen MCT 113	These are all wet wt./wet wt. 10-20% Citric Acid 3-10% Formic Acid 3-10% Hydroxyacetic acid 3-10% N-hydroxyethylenediamine triacetic acid trisodium salt	Double Walled	405	4'-0"	6'-1"	No	No	7 threaded	4	3 -Flanged 1-Flanged backup	1

Attachment #4: Table 3 Tank Accessories

Tag	Name	Overflow Tank and Size, gal**	Outlet, in	Overflow, in	Inlet, in	Fill, in	Gaskets	Metals	Manufacturer or equal	Insulation and Heating
05-05-002	UF Break Tank - Seawater	No	8 - Bolted double flange fittings	8	8 - Bolted double flange fittings	3 - BWRO reject return	EPDM	Titanium	Syndar Industries	Yes
05-07-003	CIP Tank	Yes (100 gal)	3 - Bellow	3	3 - Sidewall	Two- 3-inch openings at top of tank*	EPDM	Titanium	Syndar Industries	Yes
02-07-001	Citric Acid	Yes (50 gal)	1 - bolted flange fitting	2	N/A	1	EPDM	316SS	Polyprocess Safety Tank	No
02-07-002	Sulfuric Acid	Yes (50 gal)	1 - bolted flange fitting	2	N/A	1	Viton	C-276	Polyprocess Safety Tank	No
02-07-003	Sodium Bisulfite	Yes (75 gal)	2 - bolted flange fitting	3	N/A	2	EPDM	316SS	Polyprocess Safety Tank	No
02-07-004	Caustic Soda	Yes (75 gal)	2 - bolted flange fitting	3	N/A	2	EPDM	316SS	Polyprocess Safety Tank	No
02-07-005	Anti-scalent HYPERPERSE MDC714	Yes (50 gal)	1 - bolted flange fitting	2	N/A	1	EPDM	316SS	Polyprocess Safety Tank	No
02-07-006	Kleen MCT 405	Yes (50 gal)	1 - bolted flange fitting	2	N/A	1	EPDM	316SS	Polyprocess Safety Tank	No
02-07-007	Sodium Hypochlorite	Yes (75 gal)	2 - bolted flange fitting	3	N/A	2	EPDM	Titanium	Polyprocess Safety Tank	No
02-07-008	Kleen MCT 113	Yes (50 gal)	1 - bolted flange fitting	2	N/A	1	EPDM	316SS	Polyprocess Safety Tank	No

Notes

- * Two 3-inch openings for the following source waters: 1.) heating recirculation, and 2.) neutralization recirculation.
- ** Overflow tank will receive the overflow from the bulk tank have a vent and be provided with a 1-inch flanged connection to accept a level and accessories. The float in the tank will alarm the facility. In the case of the CIP tank, it will shut down the CIP loop and/or the plant.